PARTH THAKRE

AI Systems Engineer

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PROFESSIONAL SUMMARY

AI Systems Engineer who builds enterprise LLM systems end to end: retrieval and OCR pipelines, tool-calling agents, multi-model evaluation, and the runtime they deploy onto. Delivers client-facing automation over messy business data, including Kinaxis RapidResponse question answering, local-first AI workspaces, and Azure-based prototypes. Every answer these systems produce is traceable to its source through citations, logs, schemas, and reproducible run artifacts. Deploys on Azure, AWS, a client's own tenant, or fully on-premises, so data residency is a deployment choice rather than a constraint.

TECHNICAL SKILLS

Languages
Python, TypeScript, JavaScript, C/C++, Java, Bash

AI/LLM Systems
Azure AI Foundry, Azure OpenAI, RAG, OCR, tool and function calling, MCP, prompt and schema design

Local Inference
llama.cpp, vLLM, Ollama, LM Studio, OpenAI-compatible runtimes

Evaluation
LLM benchmarks, multi-model orchestration, structured rubrics, simulation metrics, artifact schemas

Backend/Product
FastAPI, Flask, Express, Node.js, REST APIs, SQLite, Pydantic, Zod

Frontend/Desktop
React, Next.js, Vite, Electron, HTML, CSS

Infrastructure & Cloud
Docker, Docker Compose, Linux, reverse proxies, Microsoft Azure, Azure AI Search, Microsoft Fabric, AWS

AI Coding Tools
Codex, OpenCode, OpenClaw, agentic coding workflows

Microsoft Platforms
Copilot Studio, Power Apps

DEPLOYMENT AND DATA BOUNDARY

- **Managed cloud:** Azure AI Foundry and Azure OpenAI, or AWS, in my subscription or the client's.
- **Client tenant:** deployed inside the client's own cloud account, so data and keys never leave their boundary.
- **On-premises and air-gapped:** self-hosted open-weight models through llama.cpp, vLLM, Ollama, or any OpenAI-compatible runtime.
- Applications target an OpenAI-compatible interface, so the same agent, retrieval, and citation logic runs against a hosted frontier model or a local open-weight one without a rewrite.

ENGINEERING APPROACH

- Converts AI prototypes into deployable artifacts: Electron apps, FastAPI services, Dockerized agents, CLI packages, and rendered reports.
- Designs verifiable outputs using citations, structured JSON contracts, debug traces, scorecards, and reproducible run artifacts.
- Treats retrieval, tool use, citations, and failure recovery as baseline product behavior rather than demo extras.
- Keeps client source and client data private by default; publishes only work that is mine to publish.

SELECTED EVIDENCE

- The Kinaxis pilot replaced manual worksheet navigation with natural-language questions answered from RapidResponse data, each answer carrying its workbook, worksheet, column, row, applied filters, and exact returned value.
- Evaluation projects produce inspectable outputs: four-phase tournaments, five-axis scorecards, per-call logs, and structured artifacts.
- Uses agentic coding tools for implementation, review, and debugging while retaining source control, logs, and behavioral verification.

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SELECTED AI/LLM PROJECTS

Threads
Electron, React, TypeScript, Azure AI Foundry

Local-first desktop AI workspace; private source, OSS release in progress

- Built an Azure-backed desktop AI workspace with Microsoft Entra ID and API-key authentication, memory and retrieval, and searchable local history supporting hundreds of conversations.
- Implemented text extraction for PDF, DOCX, PPTX, XLSX, CSV, Markdown, and source-code files, plus OCR for scanned PDFs and images.
- Added MCP tool discovery and selection, Azure web search grounding, Claude model routing through Azure AI Foundry, and GPT Image generation and editing workflows.

Story Tourney | GitHub
Node.js, TypeScript, Next.js, SQLite

Blind multi-model LLM evaluation platform

- Orchestrated two to four LLMs through OpenRouter across four structured phases: generation, blind peer review, revision, and final ranking.
- Added four Zod-validated JSON contracts, archived provider calls, encrypted storage for OpenRouter API keys, and a deterministic mock mode for offline testing.
- Implemented Borda count scoring with first-place and peer-review tie-breakers for repeatable tournament rankings.

Supply Docket
Python, FastAPI, Semantic Kernel, Docker

Kinaxis supply-chain query agent; client system, source stays private

- Built a locally deployed agent-and-API architecture in which Semantic Kernel tools query Kinaxis RapidResponse worksheet data through a FastAPI wrapper.
- Added eight diagnostic playbooks, catalog-aware query planning, column-ID filtering guidance, and per-call debugging metrics.
- Attached workbook, worksheet, column, row, applied-filter, and exact-value source metadata to each answer.
- Supported Azure OpenAI and local OpenAI-compatible runtimes, including LM Studio and Ollama, through configurable deployment settings.

AISB
Python, CLI, Simulation, Benchmarking

AI Society Benchmark; public release pending an in-progress refactor

- Created an LLM benchmark that simulates societies with five distinct roles, resource constraints, trust, influence, resentment, and role-specific actions.
- Refactored the benchmark into an 18-module CLI package with a canonical data layout, typed artifact schemas, and structured event logs.
- Replaced single-metric evaluation with five-axis scorecards covering welfare, economic performance, social cohesion, institutional stability, and resistance to degenerate strategies.

WORK EXPERIENCE

MaSyCoDa Solutions Pvt. Ltd.
Dec 2025–Present

AI Systems Engineer | Nagpur | the firm I deliver client engagements through

- Built enterprise AI prototypes using Azure AI Foundry, Azure OpenAI, Azure AI Search, Microsoft Fabric, and related Azure services.
- Integrated Azure AI Foundry agents with Microsoft Copilot Studio and Power Apps to create client-facing automation demonstrations.
- Deployed a Kinaxis RapidResponse supply-chain agent pilot designed to reduce manual worksheet navigation through evidence-backed question answering.
- Added eight diagnostic playbooks, citation requirements, and per-call debugging traces to make supply-chain answers reproducible and auditable.
- Used Codex, OpenCode, OpenClaw, and agentic coding harnesses to accelerate implementation, review, debugging, and iteration.

EDUCATION

Shri Ramdeobaba College of Engineering and Management

B.Tech in Computer Science and Engineering
Honors in AI and Data Science
2022–2026

PRIVATE INFRASTRUCTURE

Operates a private Linux infrastructure lab running more than 20 containers for self-hosted services, AI tooling, reverse-proxy routing, and monitoring. Uses the lab to practice deployment, privacy-aware runtime design, service recovery, and reliability engineering.